

SOFTWARE TEST PLAN

(STP)

Project: Online Bookstore

Version: 1.0

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Technology Stack: JavaScript (MERN Stack) / Python (Django)

Revision History

Version	Date	Author	Change Summary	Approval
1.0	17-09-2026	Team 4	Initial test plan for Online Bookstore	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Project Mentor			

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1. Introduction

Purpose: This document defines the testing activities planned for Online Bookstore version 1.0. It records the scope, strategy, environment, responsibilities, risks and criteria used to judge whether the application is ready for demonstration or submission.

Scope: Testing covers the customer and guest flows in the UML diagram: registration, login, browsing, searching, filtering, book details, cart, wishlist, checkout, payment, order history, tracking, cancellation and reviews. It also covers administrator functions such as books, categories, inventory, orders, users, reviews and reports.

References: Online Bookstore SRS v1.0, Customer & Guest Use Case Diagram, Administrator Use Case Diagram, project design specifications and RTM.

Definitions: SRS – Software Requirements Specification; STP – Software Test Plan; RTM – Requirements Traceability Matrix; UAT – User Acceptance Testing; API – Application Programming Interface.

2. Test Items

- Authentication module
- Catalogue, search and category filter module
- Book details and review module
- Cart and wishlist module
- Checkout, payment and order module
- Order history, tracking and cancellation
- Administrator catalogue and category management
- Inventory, user and order management
- Review moderation and sales/order reports
- Frontend, backend API and database interactions

3. Features to be Tested

The test coverage below is mapped to the requirement identifiers used in the SRS.

SRS ID	Feature	Main Test Focus
OBS-F-001 to OBS-F-003	Registration, login and logout	Valid/invalid credentials, duplicate email, protected access and logout
OBS-F-004 to OBS-F-007	Browse, search, filter and book details	Correct results, category filtering, missing books and accurate details
OBS-F-008 to OBS-F-010	Cart and wishlist	Add, update, remove, quantity limits and persistence
OBS-F-011 to OBS-F-014	Checkout, payment and orders	Address validation, totals, payment outcomes, history and cancellation
OBS-F-015	Ratings and reviews	Eligible users, validation and review visibility
OBS-F-016 to OBS-F-018	Administrator functions	Role access, catalogue, inventory, orders, users, moderation and reports
OBS-NF-001 to OBS-NF-005	Quality requirements	Response time, availability, responsive UI, errors and data integrity
OBS-SR-001 to OBS-SR-005	Security	Password protection, access control, input validation, HTTPS and safe logs

4. Features Not to be Tested

- Internal implementation of third-party payment providers beyond the available sandbox or API response.
- Physical delivery operations and courier-company internal systems.
- Actual banking settlement and refund processing outside the test environment.

- Production scaling and infrastructure monitoring beyond the project setup.
- Advanced features not included in the current SRS.

5. Test Approach / Strategy

Testing Levels

- Unit testing: individual frontend components, backend functions and validation logic.
- Integration testing: frontend, backend API, database and payment sandbox interactions.
- System testing: complete flows such as browse → cart → checkout → order.
- Acceptance testing: confirmation that the implementation satisfies the SRS and UML use cases.

Testing Types

- Functional testing of customer, guest and administrator features.
- Negative testing for invalid inputs, unavailable books, wrong credentials and failed payments.
- Regression testing after changes to authentication, catalogue, cart, payment or order modules.
- Performance testing for catalogue, search and book-detail requests.
- Usability and responsive testing on desktop and mobile screen sizes.
- Security testing for authentication, role permissions and sensitive data handling.

Entry Criteria

- A stable testable build is available.
- Database records and sample test data are prepared.
- Frontend, backend and API services can be started.
- Test accounts, books, stock values and payment sandbox details are available.

Exit Criteria

- All high-priority planned test cases are executed.
- No critical or blocker defect remains open.
- Failed cases are fixed, retested or documented as accepted limitations.
- Main customer order flow and administrator flow pass acceptance testing.
- Authentication and role-based access checks are completed.

5.1 Security Validation

- Verify that passwords are hashed and never stored in plain text.
- Check that customers cannot access administrator pages or APIs.
- Test invalid and unexpected input in login, search, cart, review and admin forms.
- Check HTTPS for sensitive communication in the deployed environment.
- Confirm that logs do not expose passwords, tokens or payment secrets.
- Test logout and session/JWT expiry behaviour where applicable.

6. Test Environment

Area	Planned Setup
Hardware	Team laptops/desktops and representative mobile screen sizes.
Operating systems	macOS, Windows or Linux as available; mobile browser testing where possible.
Software	React frontend, Node.js/Express or Django backend, database and payment sandbox.

Browsers	Google Chrome and at least one additional modern browser.
Tools	Postman, browser developer tools, Git/GitHub, manual test sheets and shared defect log.
Test data	Dummy users, books, categories, stock values, carts, orders, reviews and test payment responses.

7. Test Schedule

Dates may be adjusted according to implementation progress and the availability of a stable build.

Milestone	Planned Date / Period
Test plan preparation	17-Sep-2026
Test case design and RTM mapping	18-Sep-2026 to 20-Sep-2026
Environment and test data setup	28-Sep-2026 to 29-Sep-2026
Unit and integration testing	30-Sep-2026 to 04-Oct-2026
System and regression testing	05-Oct-2026 to 09-Oct-2026
Security and performance checks	10-Oct-2026 to 11-Oct-2026
UAT and final corrections	12-Oct-2026 to 14-Oct-2026
Test summary and sign-off	15-Oct-2026

8. Test Deliverables

- Approved Software Test Plan
- Functional and non-functional test cases
- API test collection or manual API records
- Test data and test account details
- Execution results and screenshots where required
- Defect reports
- Updated RTM
- Final Test Summary Report

9. Roles and Responsibilities

Role	Assigned To	Responsibility
QA/Test Lead	Nominated Team 4 member	Maintain plan, coordinate execution and track progress.
Test Engineers	All team members	Prepare cases, execute tests, record evidence and report defects.
Developers	All team members	Fix defects, support debugging and explain implementation behaviour.
Reviewer / Mentor	Project mentor or coordinator	Review coverage, clarify requirements and provide feedback.

10. Risks and Mitigation

Risk	Possible Effect	Mitigation
Stable build is delayed	Testing starts late	Prepare cases early and test completed modules separately.

Payment sandbox unavailable	Checkout testing is blocked	Use mocked success/failure responses and document the limitation.
Incomplete test data	Results may be unreliable	Maintain controlled dummy users, books and orders.
API/database changes	Existing tests may fail	Run regression tests after schema or API changes.
Limited testing time	Quality issues may remain	Prioritise security and the main customer/admin flows.

11. Assumptions and Dependencies

- The SRS and UML diagrams represent the agreed scope for version 1.0.
- The team will provide a testable build and access to required services.
- The database will contain enough sample data for catalogue and order testing.
- The payment gateway will provide sandbox or mock responses.
- Administrators will have separate authorised accounts.

12. Suspension and Resumption Criteria

Testing may be suspended when:

- The test environment is unavailable for a prolonged period.
- A build defect blocks a major part of the planned test cases.
- Database corruption or missing test data makes results unreliable.
- A required external service is unavailable and no substitute exists.

Testing may resume when:

- The environment is restored and smoke testing passes.
- Blocking defects are fixed or a usable build is supplied.
- Required test data and service substitutes are available.

13. Test Case Management and Traceability

Test cases will use module-based identifiers such as TC-AUTH-01, TC-CAT-01, TC-CART-01, TC-CHK-01, TC-ORD-01, TC-ADM-01 and TC-SEC-01.

The RTM will connect each SRS requirement with one or more test cases so that important requirements are not missed.

SRS Requirement	Example Test Case(s)	Coverage
OBS-F-001 to OBS-F-003	TC-AUTH-01 to TC-AUTH-03	Registration, login, logout and protected access
OBS-F-004 to OBS-F-007	TC-CAT-01, TC-SRCH-01, TC-BOOK-01	Browse, search, filtering and details
OBS-F-008 to OBS-F-010	TC-CART-01, TC-CART-02, TC-WISH-01	Cart and wishlist
OBS-F-011 to OBS-F-014	TC-CHK-01, TC-PAY-01, TC-ORD-01, TC-ORD-02	Checkout, payment and order lifecycle
OBS-F-015 to OBS-F-018	TC-REV-01, TC-ADM-01 to TC-ADM-03	Reviews and administrator functions
OBS-NF-001	TC-PERF-01	Response-time measurement
OBS-SR-001 to OBS-SR-005	TC-SEC-01 to TC-SEC-05	Security validation

14. Test Metrics and Reporting

- Percentage of planned cases executed.
- Percentage of passed, failed, blocked and not-run cases.
- Number of open defects by severity.
- Defect retest and closure status.
- Requirement coverage from the RTM.

- Number of critical or blocker defects remaining before sign-off.

Reports: The team will maintain execution updates during testing and prepare a final Test Summary Report containing results, unresolved defects, requirement coverage and readiness for demonstration or submission.

15. Approvals

Role	Name	Signature / Date
QA/Test Lead		
Development Lead		
Project Mentor		
Course Coordinator		